

Yiran Hu

Ph.D. Student in Computer Science, Purdue University
+1 765-543-3520 · hu954@purdue.edu · [Google Scholar](#) · [LinkedIn](#) · [Homepage](#)

SUMMARY

Ph.D. researcher who builds, evaluates, and optimizes LLM coding agents, spanning agent system design, agent skills, and benchmarks. First-author work on test-driven repository-level code generation (ISSRE 2026) and agent cost efficiency analysis; co-author of REPOCOD (ACL 2025) and SELP (ICRA 2025, **Best Paper Award Finalist**).

EDUCATION

Purdue University, West Lafayette, IN Aug 2023 – May 2028 (Expected)
Ph.D. in Computer Science · Advisor: [Prof. Lin Tan](#)
Tianjin University, Tianjin, China Sep 2019 – Jun 2023
B.S. in Computer Science and Technology · GPA: 89.45/100 (3.73/4.0) · Advisor: [Prof. Junjie Chen](#)

PUBLICATIONS

- [P1] **Analyzing and Mitigating Cost-Inefficient Behaviors in Coding Agents** Preprint 2026
Yiran Hu, Shanchao Liang, Nan Jiang, Yi Wu, Lin Tan.
- [C1] **TENET: One Step Toward Test-Driven Development for Repository-Level Code Generation** ISSRE 2026
Yiran Hu, Nan Jiang, Shanchao Liang, Yi Wu, Lin Tan. [\[PDF\]](#)
- [C2] **Can Language Models Replace Programmers for Coding? REPOCOD Says ‘Not Yet’** ACL 2025
Shanchao Liang, Nan Jiang, Yiran Hu, Lin Tan. [\[PDF\]](#)
- [C3] **SELP: Generating Safe and Efficient Task Plans for Robot Agents with LLMs** ICRA 2025
Yi Wu, Zikang Xiong, Yiran Hu, Shreyash S. Iyengar, Nan Jiang, Aniket Bera, Lin Tan, Suresh Jagannathan. [\[PDF\]](#)
Best Paper Award Finalist
- [C4] **LocRDF: An Ontology-Aware Key-Value Store for Massive RDF Data** WISA 2022
Yunxiao Zhang, Jinghan Li, Xueyang Liu, Rong Cheng, Yiran Hu, Xin Wang. [\[PDF\]](#)

RESEARCH EXPERIENCE

ASSET Research Lab, Purdue University – Graduate Research Assistant Aug 2023 – Present

Analyzing and Optimizing Cost Efficiency of Coding Agents [P1, first author]

- Built a trajectory-analysis pipeline over **1,200 Claude Code and Mini-SWE-Agent** trajectories on SWE-bench Verified, identifying three cost-inefficient behaviors (**subsumed retrieval, similar script generation, test re-execution**) that appear in 79%–98% of tasks and consume up to **22.75%** of task cost.
- Normalized heterogeneous agent traces into a unified action schema and combined **rule-based and LLM-based labeling** for fine-grained behavior analysis across agent architectures.
- Evaluated **structure-aware retrieval** ([CodeGraph](#) via MCP), **agent-synthesized skills**, and developer-designed skills over **10k+ agent runs** on held-out **SWE-bench Verified and Pro** tasks, executed as **12–32 parallel Docker containers** with automatic retry (~\$5K API budget), measuring Pass@1, cost, and cost-of-pass across repeated runs.
- Designed behavior-targeted **agent skills** that cut task cost by **7.88%–41.73%** and the frequency of the cost-inefficient behaviors by up to **88.91%** while maintaining Pass@1.

TENET: Test-Driven Coding Agent for Repository-Level Code Generation [C1, first author]

- Built a test-driven coding agent combining **diversity-based test selection, a retrieval/debugging toolset, and reflection-based refinement** for repository-level code generation.
- Achieved **69.08%** Pass@1 on REPOCOD and **81.77%** on RepoEval with Claude Sonnet 4, beating the strongest agent baselines (incl. OpenHands, SWE-Agent) by **9.49** and **2.17** pp (**20.10** and **6.97** pp with DeepSeek-V3), while using **3× fewer input tokens** than SWE-Agent.
- Designed a test harness that exposes only a small, diverse set of developer-written tests as executable specifications and holds out the rest for evaluation, guarding against **test overfitting and reward hacking**.

REPOCOD: Repository-Level Code Generation Benchmark [C2]

- Curated **980 tasks from 11 popular Python repositories** with an average of 314 developer-written tests per task; none of the 10 evaluated LLMs exceeded 30% Pass@1.
- Built **repository-specific Docker environments** for reproducible, **execution-based evaluation** that double as **verifiable-reward environments** for agent training; compared **RAG** strategies, including BM25.

SELP: Safe and Efficient LLM Planners for Robot Agents [C3]

- Implemented the **equivalence-voting mechanism** that improves the reliability of LLM-generated Linear Temporal Logic (LTL) specifications.
- Built two datasets for drone navigation and tabletop manipulation and contributed to **supervised fine-tuning (SFT)** and evaluation; SELP improved safety rate by 10.8% (navigation) and 20.4% (manipulation) over prior planners.

Software Analysis and Intelligence Lab, Tianjin University – Undergraduate Researcher Nov 2021 – May 2023

- Deployed the LEMON **deep-learning library testing** tool on MindSpore and applied **differential testing** to the LLVM and TVM compilers.
- Optimized ResNet-50 with TVM graph- and operator-level tuning, reducing inference latency by 14.4 ms.

SELECTED PROJECTS

Test-Time Compute Scaling for LLM Math Reasoning – CS 587 Deep Learning Spring 2026

- Led the **Self-Consistency** and **PRM-weighted Best-of-N** experiments in a team study of test-time scaling strategies (vs. Beam Search, DVTS, MCMC Power Sampling) on **MATH-500**, using Llama-3.2-1B/3B-Instruct with a Llama-3.1-8B process reward model at budgets $n=1-64$.
- Showed via oracle-vs-selected analysis that Self-Consistency leaves **25–35 pp** of accuracy unclaimed at $n=64$ (a correct answer exists in 73%–81% of candidate pools), while PRM weighting lifts 1B accuracy from 23.2% to **43.4%**.

AI Agent Security: Access Control and Memory Poisoning – CS 592 AI Agent Security Spring 2026

- Audited **Google ADK**'s tool, session, memory, and auth model against 20+ agentic **IAM** dimensions (RBAC/ABAC, task-based access, delegation, confused deputy), surfacing silent-failure risks and proposing **task-scoped capabilities** and an intent-aware policy enforcement point.
- Designed a Trusted Memory Control Plane against **persistent memory poisoning** and **prompt injection**, with provenance-typed memory, write-path quarantine, and capability-bounded tool execution.

INDUSTRY EXPERIENCE

Zhongke Wenge Science and Technology Co., Ltd., Beijing, China May 2022 – Aug 2022

Algorithm Engineer Intern, Research & Development Center

- Diagnosed PaddleOCR failure modes on COCO-Text and SVT: background interference, dense text, curved layouts.
- Integrated SAST detection with SVTR recognition (ICDAR2015), improving text detection accuracy by **27.8%**; added a recognition module for previously incompatible detectors, improving accuracy by **15.4%**.

TEACHING

Teaching Assistant, Purdue University Aug 2023 – Present

CS 176 Data Engineering in Python · CS 251/253 Data Structures & Algorithms · CS 571 Artificial Intelligence

HONORS & AWARDS

Best Paper Award Finalist , IEEE International Conference on Robotics and Automation (ICRA)	2025
Outstanding Graduate, Tianjin University	2023
First Prize, Challenge Cup National College Student Sci-Tech Competition (Tianjin University round)	2022
First Class Scholarship, Tianjin University	2021

SKILLS

Languages: Python, C++, Java

LLM Agents: Claude Code, Codex, Google ADK, Mini-SWE-Agent, OpenHands, MCP, Agent Skills

LLM & ML: Anthropic/OpenAI APIs & SDKs, Hugging Face, PyTorch, TensorFlow, Fine-Tuning (SFT), RAG

Evaluation: SWE-bench, Benchmark Design, Agent Trajectory Analysis, Agent Efficiency Analysis

Infrastructure: Docker, Linux, Git, Pytest, Tree-sitter